EV Technology Course — Module 8: Chassis, Body, Crash Safety and Lightweighting

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An EV body must carry a 300–600 kg battery low in the floor, protect it from crushing in every kind of crash, and make up for losing the engine block that once absorbed frontal impacts. This has driven the skateboard platform, structural battery packs and very large castings. This module explains the structural design, crash safety, materials and the mass trade-offs that follow.

1. Learning objectives

- Describe the skateboard platform and how the battery is integrated into the structure.
- Explain the crash load cases that matter most for EVs, especially side-pole and underbody impacts.
- Compare steels, aluminium, castings and composites for EV bodies.
- Quantify how vehicle mass affects crash energy and consumption.
- Explain EV-specific suspension, tyre and aerodynamic choices.

2. Fundamentals

- **Mass:** an EV is typically 200–400 kg heavier than the equivalent petrol car because of the battery.
- **Centre of gravity:** the battery sits low, lowering the CG and improving stability and rollover resistance.
- **Crash energy:** heavier cars carry more kinetic energy into a crash, so the structure must absorb more.
- **No engine in front:** the engine block used to be a major crash load path; EV front structures must be designed to crush in a controlled way without it.
- **The battery must not be deformed:** cell intrusion can cause short circuits and fire, so the pack is surrounded by a protective "safety cage".

3. Detailed technical explanation

Battery integration options (see battery course Module 5)

Option	Structural role of the pack	Trade-off
Pack bolted under floor (CTM/CTP)	Little; body carries loads	Simple service, some mass penalty
Structural pack	Pack frame adds torsional stiffness	Lighter body, more complex integration
Cell-to-body / cell-to-chassis	Pack lid is the floor	Lowest mass and height; hard repair

Crash load cases that drive EV design

Load case	Why it matters for EVs	Design response
Side pole impact	Pole hits between the axles, right at the pack	Strong side sills (extrusions), cross-members, gap between sill and cells
Frontal offset / full width	No engine block to absorb energy	Crush cans, longitudinal rails, front castings, subframes
Rear impact	Rear motor or pack may be close	Rear rails, protection of HV components
Underbody impact	Road debris, bollards, speed breakers	Steel or aluminium underbody shield, ground clearance
Post-crash	HV must be disconnected, no fire, rescue possible	Pyrofuse, contactor opening, isolation checks, rescue sheets

India: Bharat NCAP (launched in 2023) rates adult and child occupant protection; EVs such as the Tata Nexon EV, Punch EV and Mahindra BE 6 / XEV 9e have received 5-star ratings (check current listings). AIS-038 Rev 2 requires HV safety after crash tests.

Materials

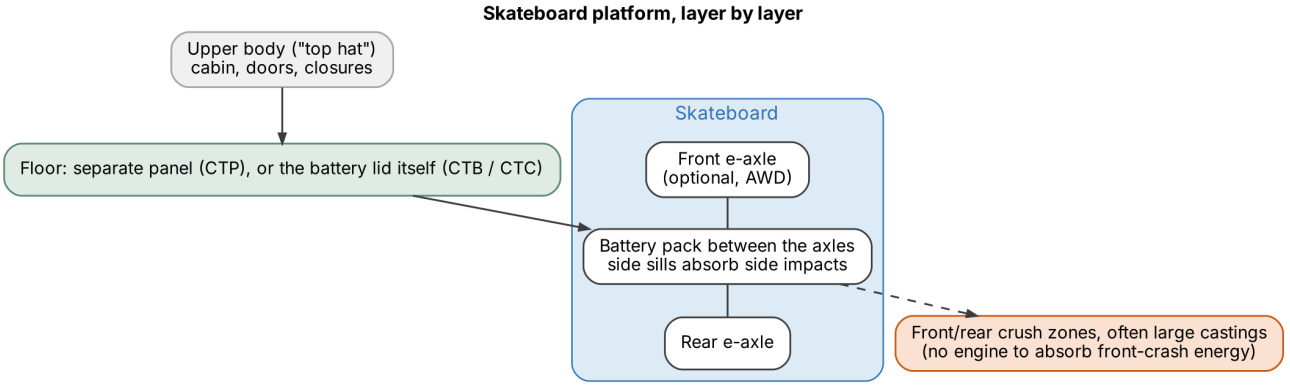
Material	Use	Strengths	Weaknesses
Advanced high-strength steel (AHSS), press-hardened steel (PHS)	Pillars, sills, cross-members	Strength at low cost; mature	Heavy relative to aluminium
Aluminium extrusions	Sills, pack frames, crash rails	Light, energy-absorbing	Cost, joining

Material	Use	Strengths	Weaknesses
Aluminium high-pressure die castings (incl. "gigacastings")	Front/rear underbodies, shock towers	Replace 50–100 parts; fewer joints	Huge presses; repair; porosity control
Composites (CFRP, SMC)	Pack lids, premium bodies	Very light	Cost, cycle time, recycling

Chassis, tyres and aero

- Springs, dampers and anti-roll bars are tuned for higher mass; bushings re-tuned because there is no engine vibration to mask road noise.
- **EV tyres:** higher load index, low rolling resistance, foam inserts for noise, compounds for high instant torque.
- **Aerodynamics:** flat underbody (the battery helps), closed grilles, aero wheels; drag coefficients around 0.20–0.30 in modern EVs.

4. Diagram: skateboard platform, layer by layer



5. Step-by-step: what happens in a side-pole crash

1. The car slides sideways into a rigid pole at about 32 km/h (typical test speed).
2. The door and side sill take the first impact; the sill (often an aluminium extrusion with internal ribs) crushes to absorb energy.
3. Cross-members and the floor carry load across to the other side of the car.
4. The pack frame resists further intrusion; a design gap between sill and cells ensures cells are not crushed.
5. The airbag controller fires the pyrofuse and opens contactors; HV is isolated within milliseconds.

6. After the test, engineers check intrusion, HV isolation, electrolyte leakage and absence of fire.

6. Engineering equations

Kinetic energy to absorb:

$$KE = \tfrac{1}{2} m v^2$$

Rolling resistance and aerodynamic drag forces:

$$F_{rr} = m g C_{rr} \qquad F_{aero} = \tfrac{1}{2} \rho C_d A v^2$$

Energy per km from a force:

$$E\ (\text{Wh/km}) = \frac{F\ (\text{N}) \times 1000\ \text{m}}{3600}$$

7. Worked examples

Example 1: crash energy. At 64 km/h (17.8 m/s, typical frontal offset test speed): EV 1,700 kg → **269 kJ**; petrol car 1,300 kg → **205 kJ**. The EV structure must absorb about 31% more energy.

Example 2: cost of extra mass. 300 kg extra with C_{rr} = 0.009 adds 26.5 N of rolling resistance → 26.5 × 1,000 ÷ 3,600 = **7.4 Wh/km** at the wheels, every kilometre, before acceleration losses.

Example 3: aero. C_d 0.28, A 2.3 m², 100 km/h: F = 0.5 × 1.2 × 0.28 × 2.3 × 27.8² = **298 N** → 83 Wh/km at the wheels. Aero dominates at highway speed.

8. Real-world examples

- **Tesla Model Y:** large front and rear castings replacing dozens of stamped parts; structural pack with 4680 cells in some versions.
- **BYD Seal:** cell-to-body (CTB) with Blade cells.
- **Xiaomi SU7, Volvo, Toyota and others:** have announced or adopted large castings.
- **Mahindra INGLO / Tata acti.ev:** dedicated skateboard architectures designed around the battery.
- **Bharat NCAP:** several Indian EVs rated 5 stars.

9. Manufacturing applications

- **Body shop:** fewer parts with castings and extrusions; more structural adhesive, flow-drill screws and self-piercing rivets for aluminium (spot welding of aluminium is harder).
- **Gigacasting:** presses of 6,000–9,000+ tonnes; die cost and maintenance; X-ray porosity checks; heat-treatment-free alloys to avoid distortion.
- **Paint shop:** aluminium parts need adapted pretreatment; heavier bodies affect conveyor ratings (your paint-shop background is directly relevant).
- **Costing:** castings shift cost from many stampings + joining labour to fewer, expensive dies and large presses; break-even depends on volume.

10. Common mistakes and misconceptions

- **"EVs are less safe because they are heavy."** Mass helps occupants in two-car crashes; low CG cuts rollover risk; the design challenge is protecting the battery.
- **"Gigacasting is always cheaper."** Only at high volume, and repair costs can rise.
- **"Aluminium means a light car."** Battery mass usually dominates.
- **"Underbody shields are optional in India."** Speed breakers and rough roads make underbody protection important.

11. Practical case study: underbody impacts on Indian roads

Problem. Field reports show scraped and dented battery enclosures from speed breakers and potholes, with some isolation warnings.

Analysis. Ground clearance under the pack lower than on the petrol twin; enclosure bottom plate thin aluminium; no sacrificial shield.

Fix. Add a replaceable steel or composite underbody shield, raise ride height slightly, add an underbody impact test (bollard/ball strike) to validation using Indian road profiles.

Lesson. Indian road conditions should be a design input, not a field surprise.

12. Revision notes

- Battery low in the floor: lower CG, more mass, more crash energy.
- Side-pole, frontal (no engine) and underbody impacts drive EV structure.
- Steels for strength at low cost; aluminium extrusions/castings for mass and part count.
- Every 100 kg costs ~2.5 Wh/km in rolling resistance alone.
- Aero dominates at highway speed.

13. Glossary

Term	Meaning
Skateboard	Flat chassis with battery between axles
Side sill	Structural member below the doors
AHSS / PHS	Advanced high-strength / press-hardened steel
Gigacasting	Very large single-piece aluminium casting
C_rr	Rolling resistance coefficient
C_d	Aerodynamic drag coefficient
Bharat NCAP	India's new car assessment programme

14. Knowledge check

Conceptual

1. Why is the side-pole test especially critical for EVs?
2. How does losing the engine change frontal crash design?
3. What are the benefits and risks of gigacasting?
4. Why do EVs need special tyres?
5. Why does battery position improve handling?

Numerical

1. Kinetic energy of a 2,000 kg EV at 50 km/h?
2. Extra rolling force and Wh/km from 200 kg with C_{rr} = 0.01?
3. Drag force at 100 km/h with C_d 0.28 and A 2.3 m²?

Application

1. Compare the costing logic for a stamped-and-welded rear underbody vs a single casting.
2. List validation tests for battery protection in an Indian EV.

15. Answer key

Conceptual

1. The pole penetrates locally between the axles, right where the battery is; there is little crush space before reaching cells.
2. Without the engine as a load path and energy absorber, front rails, crush cans and subframes must manage all the energy; front castings and careful load paths are used.
3. Benefits: far fewer parts and joints, dimensional accuracy, less tooling and labour. Risks: huge capex, die life, porosity, repair cost, alloy and heat treatment challenges.
4. Higher mass and instant torque raise load and wear; rolling resistance strongly affects range; quiet cabins make tyre noise noticeable.
5. Low CG reduces body roll and load transfer, and evenly distributed mass improves balance.

Numerical

1. 0.5 × 2,000 × 13.89² = **192.9 kJ**.
2. 200 × 9.81 × 0.01 = **19.6 N** → **5.45 Wh/km**.
3. **298 N**.

Application

1. Stamped/welded: many dies, many parts, weld lines and labour; low capex per part, flexible at low volume. Casting: one die (high cost, limited life), giant press capex, low labour, fewer fixtures; cost per part falls with volume; include scrap/porosity rate and repair cost.
2. Side pole, frontal offset and full width, rear impact, underbody impact (bollard, ball strike, speed-breaker profiles), water wading and immersion, vibration on Indian road profiles, post-crash HV isolation and no-fire checks.

16. Further learning

- Bharat NCAP protocols and results (bncap.in).
- Euro NCAP and IIHS test protocols (side pole, small overlap).
- WorldAutoSteel, *FutureSteelVehicle* and EV body studies.
- Teardown studies of Tesla Model Y castings (e.g. Munro).

Next: Module 9, brakes, steering and by-wire systems.